# Congren's Notes

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# Chapter 1

# Linux Commands

This chapter contains basic commands and advanced commands for Linux.

## 1.1 Basic Commands

### ls

```
ls # list files
ls -l # list files with permissions
ls -la # list (hidden) files with permissions
```

### touch

```
touch demo.txt # create a file
touch demo1.txt demo2.txt demo3.txt # create files
```

### mv

```
mv demo.txt ~/desktop # move a file
mv demo.txt demo1.txt # rename a file
mv demo.txt ~/desktop/demo1.txt # move and rename a file to a new place
```

### cat

```
cat demo.txt # see the contents of a file
```

### head

```
head demo.txt # check first 10 lines of the contents
head -n 20 demo.txt # check 20 lines of the contents
```

### tail

```
tail demo.txt # check last 10 lines of the contents
tail -n 20 demo.txt # check 20 lines of the contents
tail -f demo.txt # check the added contents to the file consistently
```

One example of using "tail -f" is when you want to check a log file.

### mkdir

```
mkdir demo # make a directory
mkdir -p demo1/demo2/demo3/demo4 # make a directory recursively
```

### rm

```
rm demo.txt # delete a file
rm -rf demo # delete a directory
rm -rf demo1 demo2 demo3 # delete more than one directory.
rm demo1.txt demo2.txt demo3.txt # delete more than one file
```

### cp

```
cp demo.txt ~/desktop/ # copy a file to the desktop
cp demo.txt ~/desktop/demo1.txt # copy a file and rename it
cp -r demo1 ~/desktop/demo1 # copy a directory and rename it
```

### tar

```
tar -zcvf demo.tar.gz ./demo.txt # create an archive (file)
tar -zcvf demo.tar.gz ./demo # create an archive (directory)
tar -zxvf demo.tar.gz # uncompress an archive
tar -zxvf demo.tar.gz -C ./Aaron # uncompress and rename an archive
tar -tf demo.tar.gz # see the files in an archive
```

A .tar file creates an archive without compressing
A .gz (gzip) file compresses files without creating an archive
A .tar.gz file archives and compresses multiple files into one
-z: use gzip
-c: create archives
-v: display information (not necessary to use)
-f: creates archives with given filename
-C: uncompress archives to a specific place

### find

```
find / -name demo.txt # find a file from the root
find ~/desktop -name demo.txt # find a file from the desktop
find / -name demo* # find any file that starts with demo from the root directory
find / -name *.txt # find any file that ends with .txt from the root directory
```

If you are using macOS, you have to change the command to find / -name 'demo.txt'.

## 1.2 Advanced Commands

### ps

### grep

Xie [2015]

# Chapter 2

# Matplotlib

There are three Matplotlib cheatsheets for beginners (Figure 2.1) . and intermediate users (Figure 2.2), and some tips and tricks (Figure 2.3).

https://matplotlib.org/cheatsheets/

## 2.1 Beginners

## 2.2 Intermediate users

## 2.3 Tips & Tricks

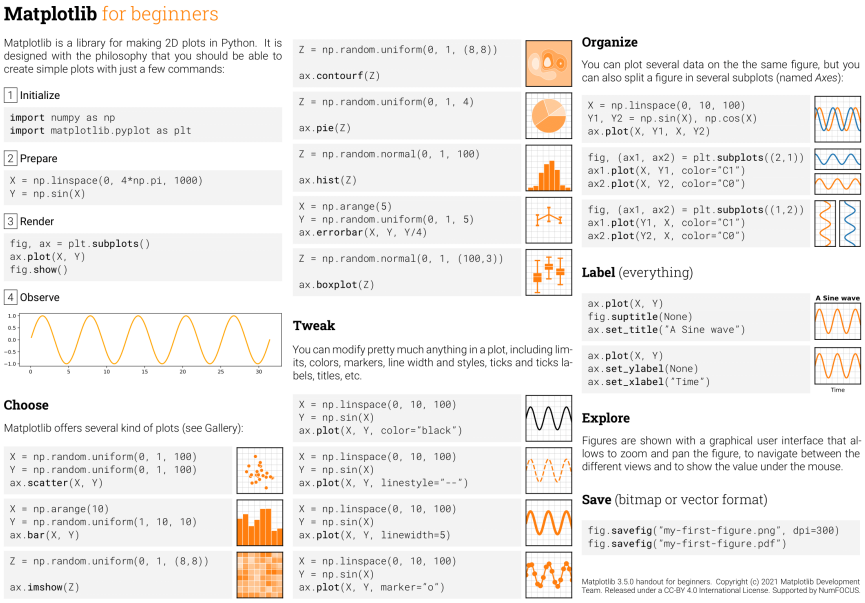


Figure 2.1: Matplotlib for beginners

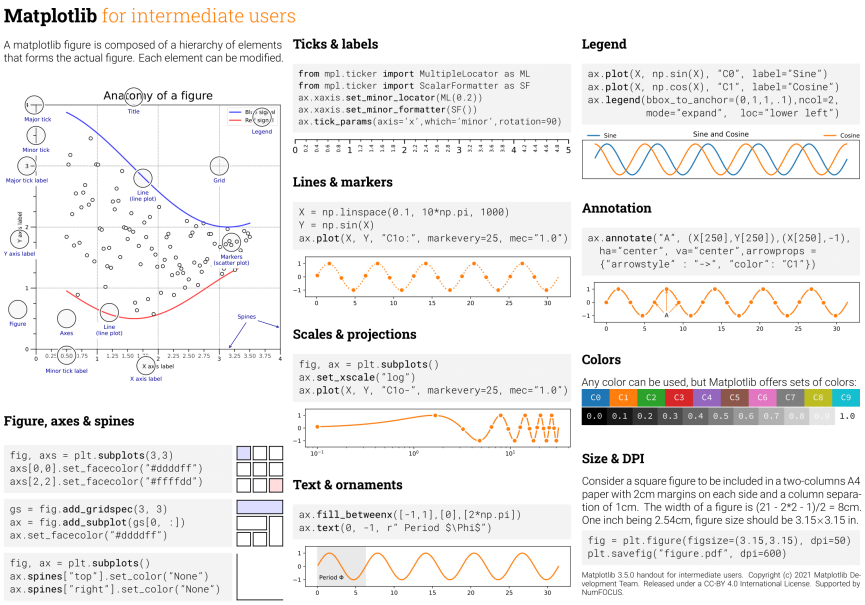


Figure 2.2: Matplotlib for intermediate users

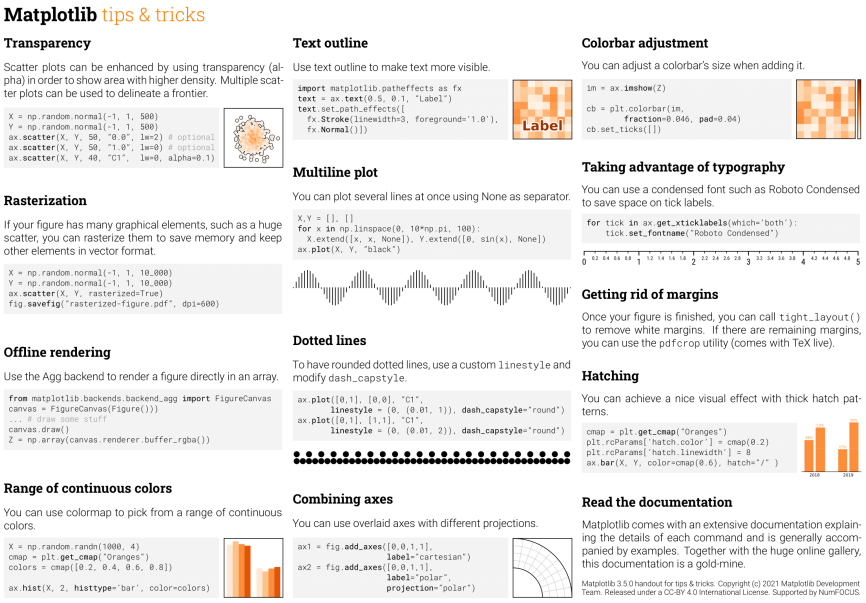


Figure 2.3: Matplotlib tips and tricks

# Bibliography

Yihui Xie. *Dynamic Documents with R and knitr*. Chapman and Hall/CRC, Boca Raton, Florida, 2nd edition, 2015. URL http://yihui.org/knitr/. ISBN 978-1498716963.